

An Improved Vulnerable Pedestrian Detection Algorithm Based on YOLOv7

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Abstract—With the spread of autonomous vehicles, vulnerable pedestrians such as the elderly and the disabled are at greater risk than ordinary pedestrians in traffic. However, due to the weak representation of vulnerable pedestrians, the existing studies paying insufficient attention to them generally received low precision and low recall. This study focuses on three deficiencies of YOLOv7, including the information loss, the weak correlation between high-level and low-level features, and the feature generalization, and proposes an improved version that can better adapt to the characteristics of vulnerable pedestrians. Firstly, by adopting the average-pooling, the enhanced spatial pyramid average-and-max pooling cross stage partial (SPAMPCSP) module can capture the global crucial information of vulnerable pedestrians, which ensures the completeness of features. Secondly, squeeze-and-excitation connection is added to realize the guidance of high-level features to low-level features and reorganize low-level features, which is conducive for classification. Thirdly, a spatial and channel efficient layer aggregation network (SCELAN) based on attention mechanism is designed to help the model focus on important features. The experiments conducted on the BG Vulnerable Pedestrian data set show that the improved YOLOv7 model reaches $mAP@0.5 = 83.08\%$, which is 4.7 higher than the original YOLOv7 model.

Keywords—vulnerable pedestrian detection, YOLOv7, SPAMPCSP module, squeeze-and-excitation, SCELAN module

I. INTRODUCTION

With the development of autonomous driving technology, how to ensure the safety of pedestrians has become an increasingly prominent problem. However, in recent studies, vulnerable pedestrians, who are more likely to be involved in accidents than average pedestrians, have received little attention.

Machine learning models proposed in the early years are mainly based on the manual labeling features, such as Haar feature [1, 2], HOG feature [3], and Deformable parts model algorithm [4], which heavily rely on prior knowledge. They are prone to overfitting in the face of complex detection and perform poorly in precision, recall, and other indicators. In addition, existing studies tend to identify vulnerable pedestrians based on their behavior. For example, Song et al. [5] provided a motion analysis module for video streams to identify vulnerable pedestrians. Liu et al. [6] proposed a guided warning system (pedestrian-oriented forewarning system) to identify pedestrians distracted by mobile phones as vulnerable pedestrians. These

behavior-based methods of identifying vulnerable pedestrians ignored naturally vulnerable groups such as the elderly, children, and the disabled, which is the common blind spot of the existing studies. Moreover, the direct application of the existing visual detectors failed to integrate the unique physical features of such vulnerable people to analyze their semantic information, which eventually leads to defects of low precision and low recall.

Neural networks in deep learning can fit very complex functions due to the rapid update of algorithms and the development of hardware. Compared with traditional machine learning algorithms based on manual annotation, it is not only more flexible in structure but also achieves higher precision and faster speed, which has won extensive attention. Among them, CNN [7] is a classic algorithm, which has become the basis for many object detection algorithms using deep learning methods. After that, the skip connection proposed by the Residual network [8] makes it possible to train deep neural networks.

The methods based on deep learning can be divided into two-stage and one-stage object detectors. The mainstream two-stage object detector is a series of algorithms based on R-CNN [9]. By generating the candidate boxes and processing pre-generated candidate regions, the two-stage object detector achieves better performance at the cost of inference time, making them less suitable for real-time detection. By making up for this shortcoming, one-stage object detectors, include YOLO series algorithms proposed by Redmon et al. [10], and the SSD algorithm proposed by Liu et al. [11], have been widely used in practice. The one-stage algorithm extracts the features of an input image and outputs the category and object position directly through an end-to-end network, which is simple and fast. But the classification precision of these methods is still unsatisfying and it can be difficult to converge when dealing with small objects, and overlapping objects recognition. Zhu et al. [12] adopted the Mask R-CNN network to achieve good precision in vulnerable pedestrian detection, but the detection time was extended. By making modifications to the YOLOv5, Devi et al. [13] improved the reliability of the detection network. The pedestrian detection method based on the YOLO network proposed by Gao et al. [14] achieved a good detection speed, but its performance in the detection of overlapping objects was poor due to the insufficient fusion of features.

In this paper, YOLOv7 [15] is selected as the baseline because it outperforms other mainstream networks. However,

due to the problems of the loss of some crucial information, the weak relation between high-level and low-level features, and insufficient fusion of long-distance semantics, the feature maps extracted by YOLOv7 tend to be incomplete and general, which may directly result in low precision and low recall. To resolve the three issues, this paper proposes an improved algorithm based on YOLOv7 for vulnerable pedestrian detection. The main contributions of this paper are summarized as follows.

- 1) Spatial pyramid average-and-max pooling cross stage partial module (SPAMPCSP) is proposed by adding multiple average pooling branches to the SPPCSP module of YOLOv7. It can supplement the object with background information, which benefits the fusion module by providing a more overall feature map.
- 2) Squeeze-and-excitation (SE) connection, a cross-stage feature guidance module, is introduced to guide low-level information with more abstract high-level information for feature recalculation.
- 3) Spatial and channel efficient layer aggregation networks (SCELAN) module is proposed by improving the original ELAN structure of YOLOv7. It fuses the semantic gap between different gradients while enhancing the feature fusion of pixel and channel.

Experimental results show the improved model reaches the mAP@0.5 of 83.08%, which is 4.76 higher than the original YOLOv7 model, and is real-time with few additional parameters.

The rest of this article is arranged as follows. Section II elaborates on the concrete improved model. Section III illustrates the data set, configuration, and experimental results. Finally, the article is summarized in Section IV.

II. METHOD

A. Improved Overall Structure

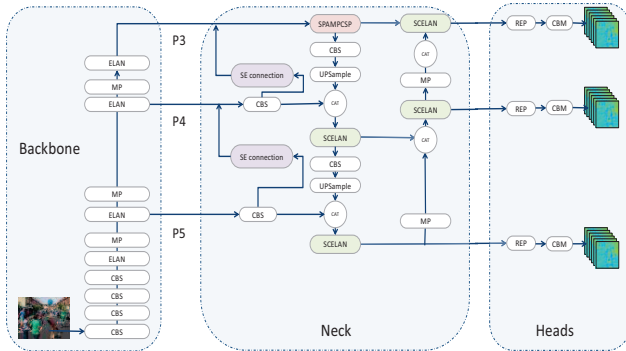


Fig. 1. Overall network architecture. The improved parts are highlighted with purple, red and green, respectively.

The improved model consists of a backbone, a neck, and three heads, which are responsible for feature extraction, feature fusion, and detection outputs, respectively. As shown in Fig. 1, through a series of convolution and pooling operations, the backbone extracts the features of the image and outputs three feature maps of different sizes P3, P4, and P5. In the neck, the SPAMPCSP module is used to make up for the loss of crucial

features. Then, two SE connections are added between the different scales of the backbone output, so that the current low-level features can be guided by the global high-level semantic information. In addition, the current branch also migrates the fully fused information to the lower-level features by employing weighted recombination. Finally, four SCELAN modules are added behind the four concatenate (CAT) operations in the neck, among which attention mechanism was applied at both cross gradient and adjacent gradient levels to fully integrate the feature maps.

B. SPAMPCSP

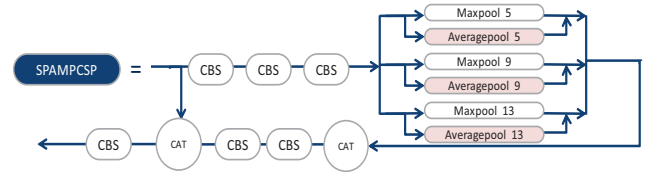


Fig. 2. SPAMPCSP module. Three average pooling (highlighted in red) were added to get more complete information of the overall data.

SPPCSP, a module used by YOLOv7 to increase the receptive field and fuse features of different sizes. It contains three branches of max pooling to obtain different receptive fields. Because max pooling tends to extract the texture information of features. However, to a certain extent, it abandons the overall information such as the background. For vulnerable pedestrian recognition, certain global features are crucial, such as guide bars and wheelchairs. Therefore, as shown in the right part of Fig. 2, the proposed SPAMPCSP module adds three average pooling branches to make up for the information loss, which helps the network to obtain more complete object information.

C. SE Connection

During the convolution process of YOLOv7, the feature maps tend to evolve into smaller sizes and more channels, which means that the data is more abstract in the spatial dimension and more informative in the channel dimension. In the process of feature extraction, there is gradually a weak correlation between high-level and low-level features as shown in Fig. 3.

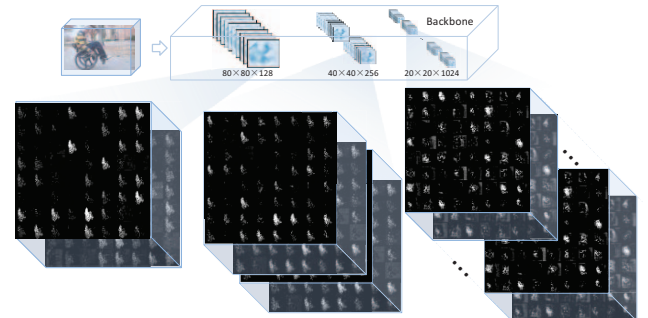


Fig. 3. Visualization of the feature map. The bottom half of the figure shows the three feature branches output by the YOLOv7 backbone network. From left to right, the levels of features gradually change from low to high.

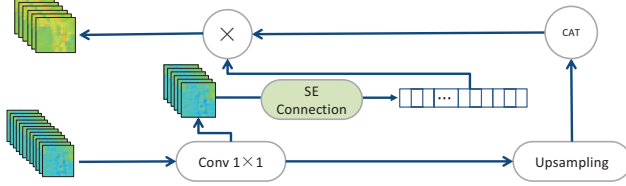


Fig. 4. Flowchart of SE connection. The process of the high-level features P4 guiding the low-level features P3 through SE connection is taken as an example. The SE connection first convolves the output P4, aligning its channel with P3. In this process, the rich channel information of P4 is condensed. Then, the channel attention of $1 \times 1 \times c$ is generated by the squeeze and inspire operation based on the SE attention mechanism for processing channel information. Then, the channel attention is multiplied by the original feature P3 to achieve the guidance of the high-level feature to the low-level feature on the channel.

- 1) From the perspective of spatial dimension, high-level information is abstract. On the contrary, low-level information contains more specific information on outline and texture and involves some noises.
- 2) From the perspective of channels, advanced information not only has more channels but also has sparser distribution among different channels, which implies that rich and abstract channel information can be exploited. The low-level features are characterized by fewer channels, dense distribution, and small distances.

Based on the two observations above and inspired by [16], it can be seen that high-level information can play a guiding role in the process of fusion with low-level information, which makes the distribution of low-level information sparser and more conducive to object recognition. According to this idea, this paper designs SE connection [17, 18], which is known for its advantage of improving features in different channels and positions, to adjust feature maps of different sizes and reorganize low-level features. The process is shown in Fig. 4.

D. SCELAN

The original ELAN module extracts features through two branches of different times of convolution with the residual idea. However, the information extracted from the network tends to generalize the features rather than highlight the features that have key support for the object detection results. Therefore, a SCELAN module as shown in Fig. 5 is designed, which contains two improvements.

- 1) Features of different depths are fused by SE connection.
- 2) Spatial attention is integrated into convolution modules in a sequential stack, which prevents the excessive semantic gap from bringing some negative effects.

III. EXPERIMENTS

A. Data Set and Configuration

The experimental data set adopted the vulnerable pedestrian detection data set (BGVP) [19], which was released by the Computer Science Department of Bowling Green State University in 2022. The data set classifies vulnerable pedestrians into four categories:

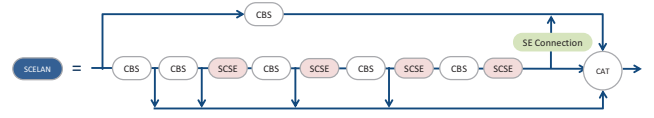


Fig. 5. SCELAN structure. Improvements include SE connection and SCSE modules (highlighted in green and red).

1) *Children without disability*: Children are considered vulnerable pedestrians because they act erratically and may not understand traffic laws. In this data set, this category includes any pedestrian between the ages of 1 and 16.

2) *Elderly without disability*: This category is defined as people aged 50 years or older without disabilities.

3) *With disability*: The most vulnerable groups include pedestrians with physical impairments who may need some form of assistance. Pedestrians can be any age group, but if they have a physical disability, they fall into this category because their age-related limitations are overshadowed by the disability. In this data set, assistive devices for the disabilities include wheelchairs, walkers, guide poles, etc.

4) *Non-vulnerable*: This category of pedestrians does not fall into any of these categories and is less vulnerable to traffic conditions than other categories.

The data set included 2,000 images with a total of 5,932 border notes. Each image has an average of three annotations, and the median image size is 600×408 pixels. “Children without Disability” is the most prominent vulnerable pedestrian category with 1,646 boundary box notes, the category of “Elderly without Disability” has 815 boundary box instances, the category of “with Disability” has 942 instances, and the category of “Non-Vulnerable” has 2,529 instances. In this experiment, the data set was divided into 1620 images as the training set, 180 images as the validation set, and 200 images as the test set.

Experiments are conducted on a computer with an NVIDIA RTX A5000 GPU and an Intel Xeon Platinum 8350C CPU. The programming language is Python 3.8.11. The CUDA version is 11.8. The picture size of the training and test is 640×640 . The whole network is optimized by the SGD optimizer with an initial learning rate of 0.01. The batch size is 8 and the number of iterations is 300.

B. Comparative Experiments

The experimental conditions here are consistent with those described above. Due to the real-time requirements of vulnerable pedestrian detection, faster R-CNN with higher speed is chosen to compare with one-stage algorithms. Specific experiments include the measurement of input size, params, GFLOPs, and mAP@0.5. The test results are shown in Table I.

TABLE I. COMPARISON BETWEEN MAINSTREAM VISUAL DETECTORS

Model	Input Size	Params(M)	GFLOPs(G)	mAP@0.5(%)
Faster R-CNN	640×640	42.00	180.00	73.32
YOLOv5-s	640×640	7.20	16.90	70.47
YOLOx-s	640×640	8.97	26.81	72.85
YOLOv7-L	640×640	37.21	105.17	78.66
YOLOv8-L	640×640	43.69	165.74	73.97

Compared with YOLOv5 [20] and YOLOx [21], YOLOv7 has made improvements in network aggregation and convolution reparametrization, which makes it more capable of semantic fusion and understanding. YOLOv8 [22] is the latest algorithm of the YOLO series. In the backbone of YOLOv8, downsampling is achieved only by convolution. In contrast to the combination of convolution and max pooling in YOLOv7, YOLOv8 has the risk of information loss in key areas. All in all, YOLOv7 outperforms other models in terms of detection performance. Therefore, this paper chooses YOLOv7 as the baseline network for improvement.

Table II shows the comparison between the three improved models and the original version of YOLOv7 under the same experimental conditions. It can be seen that the three improved models have an increase of mAP@0.5 in the categories of the “Elderly without Disability” and the “with Disability”, with an average increase of 7.78 and 2.04. In the categories of “Children without Disability” and the “Non-Vulnerable”, mAP@0.5 have an average decrease of 0.83 and 0.54. With improvements, the model pays more attention to the pose features of objects. But it doesn't work well for postural pedestrians, such as the “Children without Disability” and the “Non-Vulnerable”.

TABLE II. EXPERIMENTAL RESULTS OF THE YOLOv7 AND ITS COMBINATION WITH A CERTAIN IMPROVED MODULE

Model	mAP	AP(children without disability)	AP(elderly without disability)	AP(non vulnerable)	AP(with disability)
YOLOv7-L	78.32	86.64	57.94	75.28	93.43
+ SPAMPCSP	79.69(↑1.37)	86.35(↓0.29)	63.49(↑5.55)	74.89(↓0.39)	94.03(↑0.6)
+ SE Connection	80.38(↑2.06)	85.62(↓1.02)	65.9(↑7.96)	74.08(↓1.2)	95.94(↑2.51)
+ SCELAN	81.23(↑2.91)	85.45(↓1.19)	67.77(↑9.83)	75.24(↓0.04)	96.44(↑3.01)

C. Ablation Experiments

TABLE III. EXPERIMENTAL RESULTS OF THE YOLOv7 AND ITS COMBINATION WITH DIFFERENT IMPROVED MODULES

	YOLOv7-L	+ SPAMPCSP	+ SPAMPCSP + SE Connection	+ SPAMPCSP + SE Connection + SCELAN
Input Size	640×640	640×640	640×640	640×640
Params(M)	37.21	37.99	38.33	39.23
GFLOPs(G)	105.16	105.79	106.11	107.81
FPS	61.9	61.08	60.13	37.9
mAP(%)	78.32	79.69(↑1.37)	81.29(↑1.60)	83.08(↑1.79)
AP(%) (children without disability)	86.64	86.35(↓0.29)	83.63(↓2.72)	82.35(↓1.28)
AP(%) (elderly without disability)	57.94	63.49(↑5.55)	68.61(↑5.12)	75.68(↑7.07)
AP(%) (non vulnerable)	75.28	74.89(↓0.39)	78.33(↑3.44)	76.14(↓1.19)
AP(%) (with disability)	93.43	94.03(↑0.60)	94.57(↑0.54)	98.16(↑3.59)

The experimental conditions here are consistent with those described above. Specific experimental results include AP@0.5, mAP@0.5, and FPS. The specific test results are shown in Table III. It can be observed that mAP@0.5 of the improved model is 4.76 higher than YOLOv7.

D. Visualization and Analysis of Detection Results

The detection results of the models on the test set are visualized in Fig. 6. From Fig. 6 (1)-(2), the model identifies three false negative targets and corrects one misclassified object.



Fig. 6. Comparison of test results. Yellow circles represent the detection differences between different models.

Three false negative targets and corrects one misclassified object. Fig. 6 (3) detects a false negative object based on Fig. 6 (2). In Fig. 6 (4), almost all objects are detected. In addition, combined with Fig. 6 (5)-(8), it can be seen that the proposed model greatly improves the difficult detection problems such as small and medium-sized object detection and overlapping object detection in complex real scenes. According to Table III, the precision of “Elderly without Disability” is not high, which can be further analyzed in conjunction with Fig. 6 (9)-(10). It can be seen that the improved model detects some false negative objects relative to YOLOv7 but misjudges them as non-vulnerable groups. From Fig. 6 (10)-(11), the model completes the prediction and localization. From Fig. 6 (11)-(12), the model locates the object more accurately. It can be seen from Fig. 6 (13)-(16) that the model corrects the objects misclassified by YOLOv7. Taking the above two visualization processes as examples, it can be seen that the detection performance of the model in the category of “Elderly without Disability” has been improved.

But there is still the risk of misclassification when using the improved model. For example, the elderly people without disabilities in Fig. 6 (3) are misclassified as non-vulnerable. One of the reasons is that the limited samples of “Elderly without Disability” may lead to the problem of insufficient sample representation. Another possible reason is that the sample diversity of this category is too large to extract features effectively, which still needs further investigation.

IV. CONCLUSION

In this paper, an improved detector based on the YOLOv7 model for vulnerable pedestrians is proposed to solve the problem of information loss, a weak correlation between high-level and low-level features, and feature generalization. Firstly, by adding multiple branches of average pooling, the SPAMPCSP module extracts more representative and complete features. In addition, with the guidance of high-level features, the SE connection enhances the correlation between low-level and high-level features. Finally, a SCELAN module is designed to highlight features that have critical support for the object detection results by using channel and spatial attention mechanisms. Under the premise of real-time detection, the improved model is 4.76 higher than the original YOLOv7 on mAP@0.5, which outperforms other vision detection models.

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